

Date: 29-8-2026

Lab # 01 INTRODUCTION TO PYTHON AND SPYDER

OBJECTIVES:**Objective:**

The objective of this lab is to introduce you to Python programming and familiarize you with the Spyder IDE (Integrated Development Environment). By the end of this lab, you will be able to write basic Python programs, including printing text and performing basic arithmetic operations.

Introduction to Python:

Python is a high-level, general-purpose programming language known for its simplicity and readability. It supports multiple programming paradigms, including procedural, objectoriented, and functional programming. Python is widely used for web development, data analysis, machine learning, automation, and more.

Key Features of Python:

- Easy to learn and use due to its simple syntax.
 - Extensive libraries and frameworks available for various applications. □
Dynamic typing, meaning you don't need to declare variable types. □
Interpreted language, so code is executed line by line.
-

Introduction to Spyder:

Spyder is an open-source, cross-platform IDE specifically designed for scientific computing and data analysis. It provides powerful tools for Python development and offers features like code introspection, an integrated debugger, and a variable explorer.

Steps to Install Spyder:

1. Install Anaconda (which includes Spyder) from [here](#).
2. Launch Spyder from the Anaconda Navigator.
3. You are now ready to begin writing Python code!

Basic Python Syntax:

1. Printing Output:

The `print()` function is used to display output in Python. Let's try printing a simple message like your name and roll number.

```
python Copy
code
# Printing Name and Roll Number
print("Name: John Doe") print("Roll
Number: 12345")
```

2. Variables:

In Python, you don't need to declare the type of a variable explicitly. Let's define two variables: `name` and `roll_num`.

```
# Defining variables name
= "John Doe" roll_num =
12345

# Printing variables
print("Name:", name) print("Roll
Number:", roll_num)
```

3. Basic Arithmetic Operations:

You can use Python to perform basic arithmetic operations like addition, subtraction, multiplication, and division.

```
python Copy
code
# Basic Arithmetic Operations a
= 10
b = 5

# Performing operations
sum_result = a + b diff_result
= a - b product_result = a * b
division_result = a / b

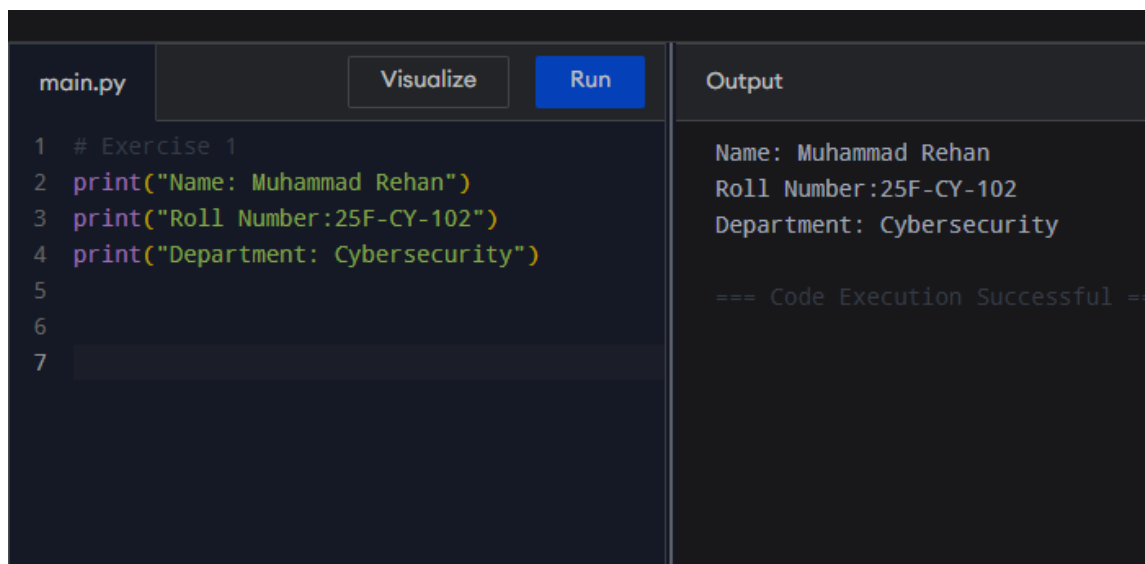
# Printing results print("Sum:",
sum_result) print("Difference:",
```

```
diff_result) print("Product:",  
product_result) print("Division:",  
division_result)
```

Exercise

1. Write a Python program to print your name, roll number, and department.
2. Create two variables, x and y, and assign them values. Perform addition, subtraction, multiplication, and division on these two variables and print the results.
3. Write a Python program that calculates the area of a circle.
4. Write a Python program to find the square and cube of a number provided by the user.
5. Write a Python program that takes two numbers as input from the user and prints their sum, difference, and product.
6. Create a Python program that accepts the marks of 3 subjects as input and calculates the total and average marks.
7. Write a program that asks the user for a number and checks if it is even or odd.

Task 1:



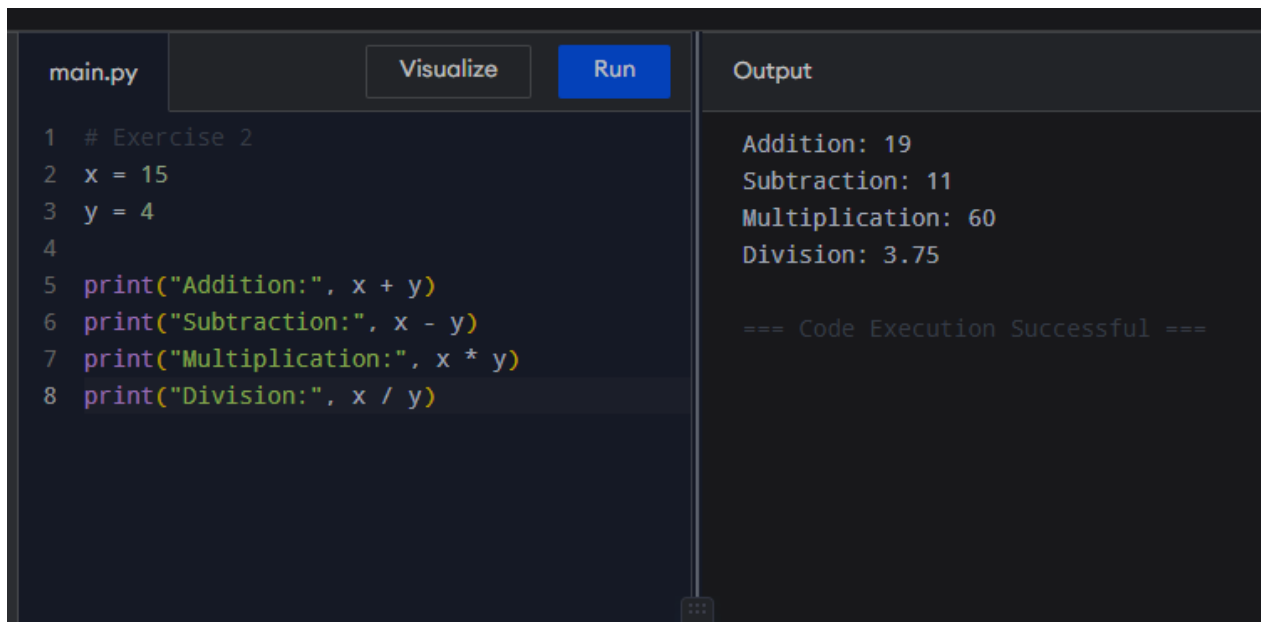
The screenshot displays a Python IDE interface. On the left, a code editor window titled 'main.py' contains the following Python code:

```
1 # Exercise 1  
2 print("Name: Muhammad Rehan")  
3 print("Roll Number:25F-CY-102")  
4 print("Department: Cybersecurity")  
5  
6  
7
```

At the top of the editor, there are buttons for 'Visualize' and 'Run'. The 'Run' button is highlighted in blue. On the right side of the IDE, an 'Output' window shows the results of the program execution:

```
Name: Muhammad Rehan  
Roll Number:25F-CY-102  
Department: Cybersecurity  
  
=== Code Execution Successful ===
```

Task 2:



The screenshot shows a Python IDE with a file named 'main.py'. The code in the editor is as follows:

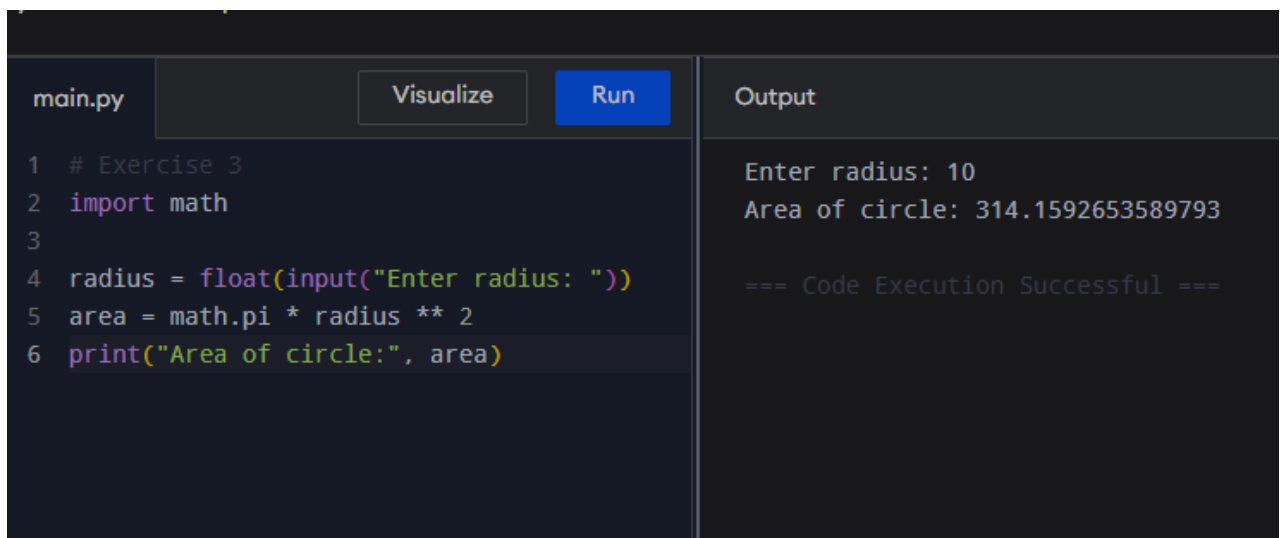
```
1 # Exercise 2
2 x = 15
3 y = 4
4
5 print("Addition:", x + y)
6 print("Subtraction:", x - y)
7 print("Multiplication:", x * y)
8 print("Division:", x / y)
```

The 'Run' button is highlighted in blue. The 'Output' pane on the right displays the results of the execution:

```
Addition: 19
Subtraction: 11
Multiplication: 60
Division: 3.75

=== Code Execution Successful ===
```

Task 3:



The screenshot shows a Python IDE with a file named 'main.py'. The code in the editor is as follows:

```
1 # Exercise 3
2 import math
3
4 radius = float(input("Enter radius: "))
5 area = math.pi * radius ** 2
6 print("Area of circle:", area)
```

The 'Run' button is highlighted in blue. The 'Output' pane on the right displays the results of the execution:

```
Enter radius: 10
Area of circle: 314.1592653589793

=== Code Execution Successful ===
```

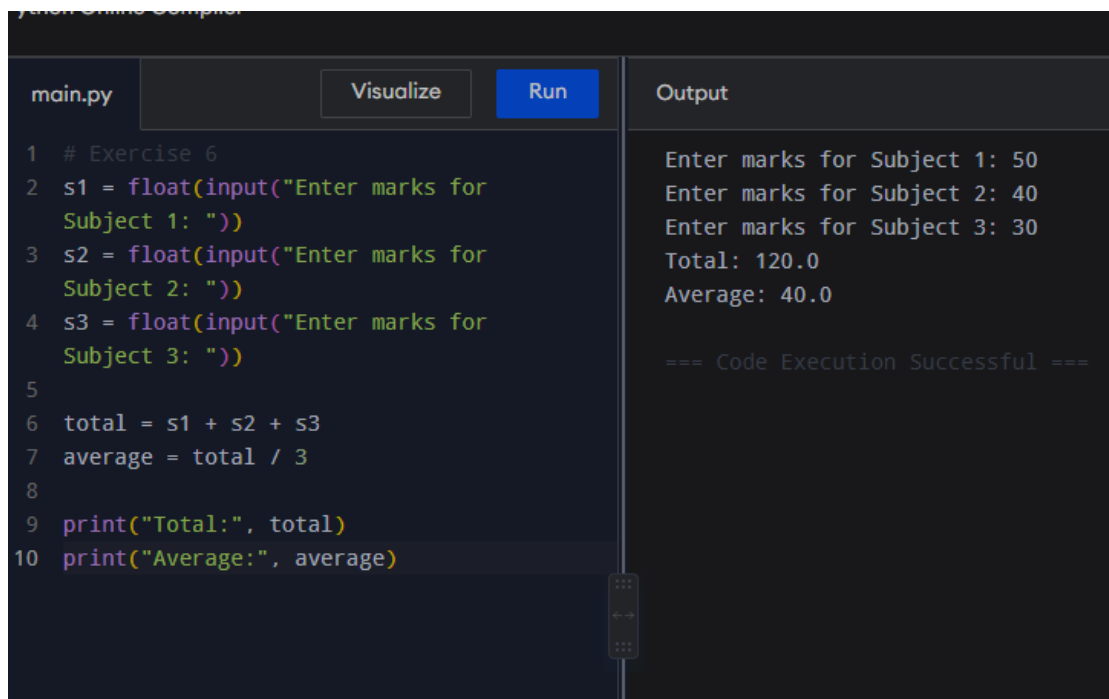
Task 4:

main.py	Visualize	Run	Output
<pre>1 # Exercise 4 2 num = float(input("Enter a number: ")) 3 print("Square:", num ** 2) 4 print("Cube:", num ** 3)</pre>			<pre>Enter a number: 5 Square: 25.0 Cube: 125.0 === Code Execution Success</pre>

Task 5:

main.py	Visualize	Run	Output
<pre>1 # Exercise 5 2 a = float(input("Enter first number: ")) 3 b = float(input("Enter second number: ")) 4 5 print("Sum:", a + b) 6 print("Difference:", a - b) 7 print("Product:", a * b)</pre>			<pre>Enter first number: 6 Enter second number: 7 Sum: 13.0 Difference: -1.0 Product: 42.0 === Code Execution Successful ===</pre>

Task 6:

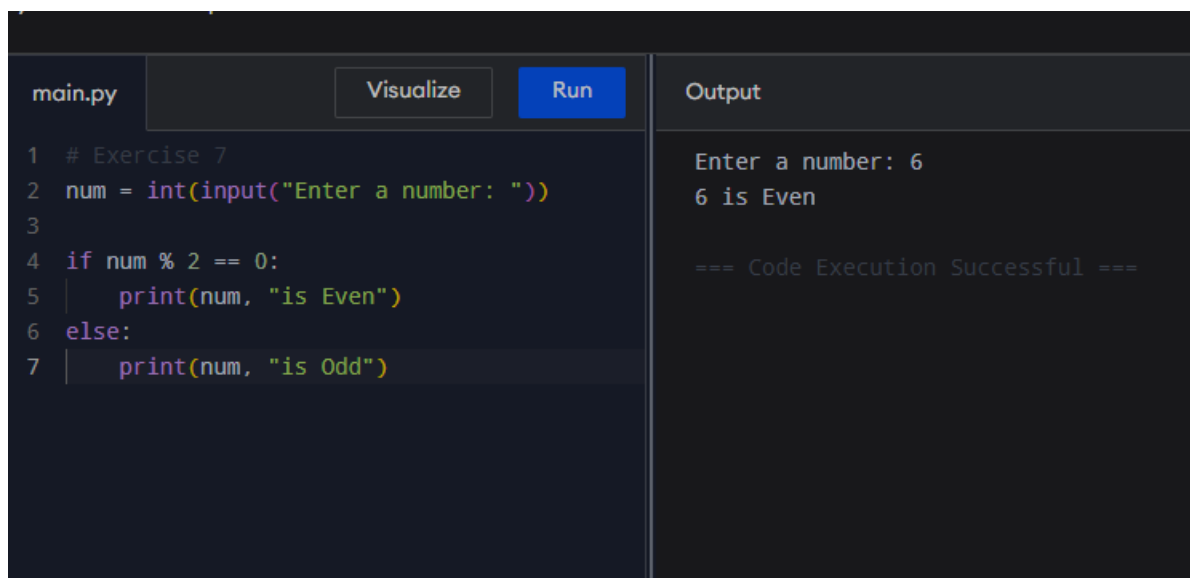


```
main.py Visualize Run Output
1 # Exercise 6
2 s1 = float(input("Enter marks for
  Subject 1: "))
3 s2 = float(input("Enter marks for
  Subject 2: "))
4 s3 = float(input("Enter marks for
  Subject 3: "))
5
6 total = s1 + s2 + s3
7 average = total / 3
8
9 print("Total:", total)
10 print("Average:", average)
```

Enter marks for Subject 1: 50
Enter marks for Subject 2: 40
Enter marks for Subject 3: 30
Total: 120.0
Average: 40.0

=== Code Execution Successful ===

Task 7:



```
main.py Visualize Run Output
1 # Exercise 7
2 num = int(input("Enter a number: "))
3
4 if num % 2 == 0:
5     print(num, "is Even")
6 else:
7     print(num, "is Odd")
```

Enter a number: 6
6 is Even

=== Code Execution Successful ===